Practice workbook · Cloud storage: what it is, pros, cons and security

What cloud storage is, its benefits and drawbacks, and its security issues including how protection duties are split between user and provider, from 1-mark items to a 6-mark discussion.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-08

1. Question 2 · part b.ii · 2 marks

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Worked-example retry · Attempt part b.ii only.

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2 Computer systems use both hardware and software.

(a) Give **two** output devices found on smartphones.

(2)

1 .....

.....

2 .....

.....

(b) Computer systems use both primary and secondary storage.

(i) Identify which **one** of these uses storage space on the hard drive as additional memory.

(1)

- ☒ **A** Cache memory
- ☒ **B** Random-access memory
- ☒ **C** Read-only memory
- ☒ **D** Virtual memory

(ii) Computer systems can use cloud storage.

Describe what is meant by **cloud storage**.

(2)

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1. Question 2 · part b.ii · 2 marks

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Worked-example retry · Attempt part b.ii only.

(c) The CPU performs the fetch-decode-execute cycle.

(i) Give **three** factors that affect the performance of the CPU.

(3)

1 .....

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2 .....

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3 .....

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(ii) Describe how the address bus is used during the fetch-decode-execute cycle.

(2)

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(d) Classical computers store data using bits that exist in one of two binary states, 1 or 0.

Quantum computers store data in a superposition of states.

Give **two other** characteristics of quantum computers.

(2)

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2 .....

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(Total for Question 2 = 12 marks)

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1. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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2. Question 2 · part a · 2 marks

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Further practice · Attempt part a only.

2 Zafer and Robert work for a company that makes washing machines.

(a) Zafer writes user manuals for the washing machines.

He stores these documents in the cloud.

Zafer and the cloud storage provider share responsibility for data security.

State **one** area of responsibility for each of them.

(2)

Zafer

.....

.....

Cloud storage provider

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.....

(b) Robert programs the robots that make the parts for the washing machines.

Zafer writes the user manuals.

Robert and Zafer have different levels of access to folders and files on the company network.

The different levels of access are Read, Write, Execute and None.

Complete the table to show the levels of access that each has.

(2)

	Robert	**Zafer**
Washing machine design drawings		
A folder of manuals for new machines		

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2. Question 2 · part a · 2 marks · Working and answer

Working and answer

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3. Question 3 · part b.ii · 2 marks

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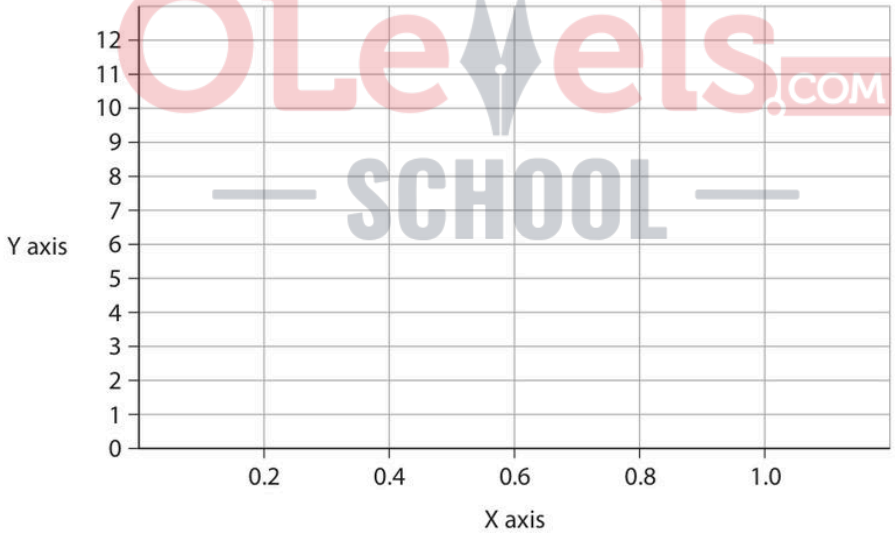
Further practice · Attempt part b.ii only.

3 Alyssa is a music producer.

(a) **Figure 1** shows the denary values of five samples of an analogue sound using a sample interval of 0.2 seconds.

Sample number	Denary value
1	1
2	10
3	12
4	5
5	3

Figure 1



(i) Complete this graph using the sample information from **Figure 1** to show the digital sound wave.

(3)

(ii) Give a suitable label for the X axis.

(1)

(iii) Give a suitable label for the Y axis.

(1)

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P 7 2 5 3 8 A 0 8 2 0

3. Question 3 · part b.ii · 2 marks

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Further practice · Attempt part b.ii only.

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(b) Alyssa uploads music files to her cloud storage.

(i) She compresses the files before she uploads them using a lossless algorithm.

Give **one** disadvantage of using a lossless rather than a lossy algorithm for this purpose.

(1)

.....

.....

(ii) Explain **one** benefit to Alyssa of storing her music files in the cloud.

(2)

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(iii) Give **one** possible security issue associated with storing music files in the cloud.

(1)

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(iv) One of Alyssa's music files is stored at https://www.cloudisfab.com/re12/ru2.mp3

Complete the table by adding a description of each URL component.

(4)

URL component	Description
https	
www.cloudisfab.com	
re12	
ru2.mp3	

(Total for Question 3 = 13 marks)

3. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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4. Question 3 · part b.iii · 1 mark

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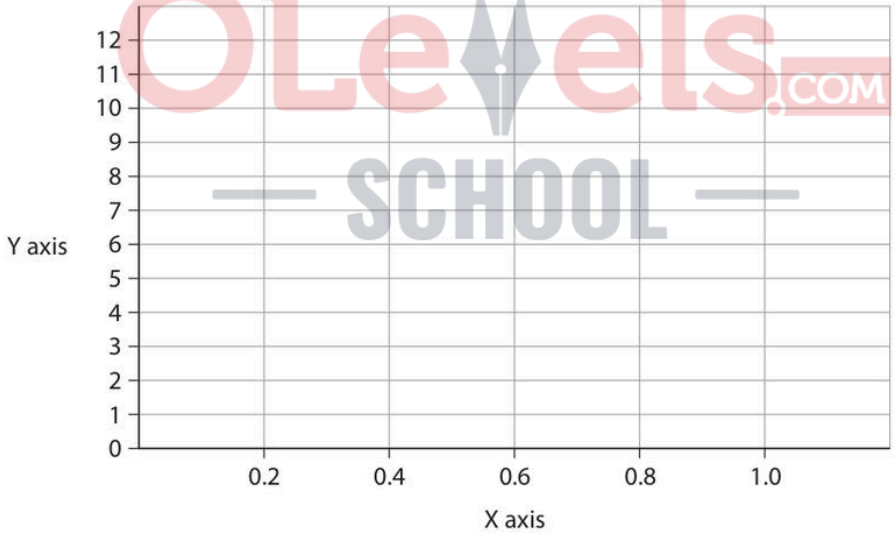
Further practice · Attempt part b.iii only.

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Figure 1



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(1)

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(1)

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8

P 7 2 5 3 8 A 0 8 2 0

4. Question 3 · part b.iii · 1 mark

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Further practice · Attempt part b.iii only.

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(b) Alyssa uploads music files to her cloud storage.

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(1)

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(2)

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(1)

.....

.....

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Complete the table by adding a description of each URL component.

(4)

URL component	Description
https	
www.cloudisfab.com	
re12	
ru2.mp3	

(Total for Question 3 = 13 marks)

4. Question 3 · part b.iii · 1 mark · Working and answer

Working and answer

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5. Question 6 · part d · 6 marks

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Further practice · Attempt part d only.

6 Programmers share algorithms with different people and write algorithms for different reasons.

(a) A programmer is showing a new algorithm to a group of non-technical managers.

State an appropriate method for writing the algorithm.

Justify your answer.

(2)

Method

Justification

(b) **Figure 1** shows an algorithm that displays a string based on the number input by the user.

```
1  SEND ("Enter a number: ") TO DISPLAY
2  RECEIVE inNum FROM (INTEGER) KEYBOARD
3  IF ((inNum = 1) OR (inNum = 2)) THEN
4      IF (inNum = 1) THEN
5          SEND ("First") TO DISPLAY
6      ELSE
7          IF (inNum = 2) THEN
8              SEND ("Second") TO DISPLAY
9          END IF
10     END IF
11 ELSE
12     SEND ("Invalid input") TO DISPLAY
13 END IF
```

Figure 1

Give **one** reason why the selection statement on line 7 is not required.

(1)

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5. Question 6 · part d · 6 marks

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Further practice · Attempt part d only.

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(c) **Figure 2** shows an algorithm that manipulates arrays.

The algorithm works with any number of scores.

```
1 SET oldScores TO [10, 20, 30, 40, 50]
2 SET newScores TO [0, 0, 0, 0, 0]
3 SET newIndex TO 0
4
5 FOR oldIndex FROM (LENGTH (oldScores) - 1) TO 0 STEP -1 DO
6     SET newScores[newIndex] TO oldScores[oldIndex]
7     SET newIndex TO newIndex + 1
8 END FOR
```

Figure 2

(i) Describe what happens to the variable **oldIndex** when line 5 is executed.

(2)

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(ii) State the purpose of the algorithm in **Figure 2**.

(1)

5. Question 6 · part d · 6 marks

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Further practice · Attempt part d only.

(d) Cloud storage has become an increasingly popular service.

Discuss the benefits and drawbacks of using cloud storage.

(6)

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(Total for Question 6 = 12 marks)

5. Question 6 · part d · 6 marks · Working and answer

Working and answer

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